

Everything You Always Wanted to Know About Momentum

Mechanics, Flavours, Crisis Alpha, and AI Frontiers

Quantitative Trading Lecture Series

September 13, 2026

Lecture Agenda

- ▶ **Introduction: Why Momentum is Everywhere**
- ▶ Two Flavours of Momentum: TSMOM vs. XSMOM
- ▶ Signal Formulations & Filter Equivalence
- ▶ Multi-Century Persistence (200-Year Proof)
- ▶ Asymmetric Tail Risks & Positive Strategy Skewness
- ▶ Factor Momentum & Macroeconomic Drivers
- ▶ The Dark Side: Momentum Crashes & Risk Scaling
- ▶ Classical vs. Adaptive State-Space Filtering
- ▶ Deep Learning Frontiers: DMNs & Transformers
- ▶ Mathematical Skewness & Convexity Proofs
- ▶ Algorithmic Grounding & Academic Bibliography

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Why Momentum is Everywhere: The Institutional Pitch

Why Pension Funds & Allocators Love CTAs

Momentum is one of the most heavily traded systematic products globally. Pension fund advisors routinely allocate to trend strategies because they offer a rare institutional property: **Tail risk protection without paying an option premium.**

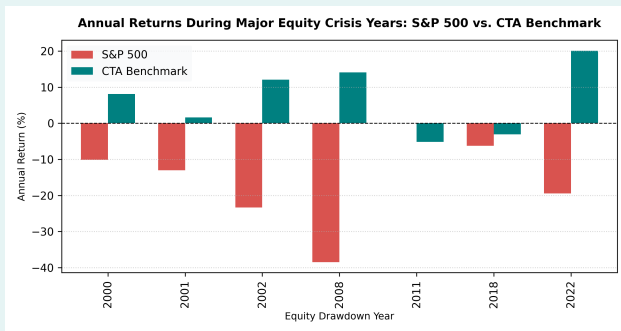


Figure: Annual returns during major equity crisis years: S&P 500 vs. CTA benchmark.

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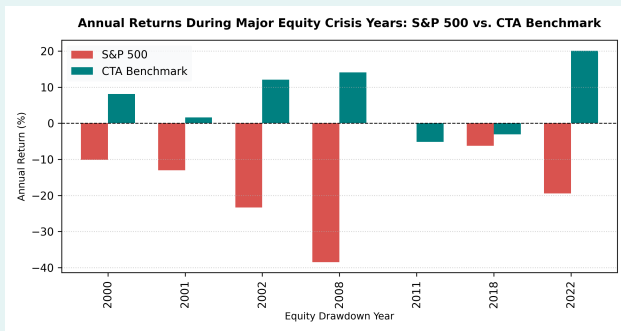


Figure: Annual returns during major equity crisis years: S&P 500 vs. CTA benchmark.

Two Flavours of Momentum: TSMOM vs. XSMOM

Time-Series Momentum (TSMOM / Absolute Trend)

- ▶ **Mechanism: Evaluates an asset against its own price history** ($s_{i,t}^{\text{TS}} = \text{sign}(r_{i,t}^{(k)})$).
- ▶ **Domain: Dominant in CTAs & Managed Futures** across liquid futures (rates, FX, commodities, equity indices). Moskowitz, Ooi, Pedersen (2012).

Cross-Sectional Momentum (XSMOM / Relative Strength)

- ▶ **Mechanism: Ranks assets within a peer universe $\mathcal{U}$, going long top quantile Q_H and short bottom quantile Q_L .**
- ▶ **Domain: Dominant in Equities Quantitative Factor Investing.** Asness, Moskowitz, Pedersen (2013).

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Signal Formulations & Filter Equivalence

Common Trend Filter Formulations

- ▶ SMA Crossover: $s_t = \text{sign}(\text{SMA}_t(k_1) - \text{SMA}_t(k_2))$
- ▶ EWMA Crossover: $s_t = \text{sign}(\text{EWMA}_t(\lambda_1) - \text{EWMA}_t(\lambda_2))$
- ▶ Donchian Channel: $s_t = +1$ if $P_t = \max_k P_{t-k}$, -1 if $P_t = \min_k P_{t-k}$

The Filter Equivalence Theorem

All linear digital trend filters can be represented as a convolution kernel $s_{t-1} = \sum_{k=1}^{\infty} \phi_k r_{t-k}$. As shown by Hamdan et al. (2012) and Bruder & Gaussell (2011), different trend filters generate positions with > 0.90 mutual correlation because all moving average filters are linear combinations of autocorrelation lag weights.

Two Centuries of Trend Following

Multi-Century Evidence (Lempérière et al. 2014)

Empirical analysis across 200 years of market data confirms that time-series momentum has generated a stable annualized Sharpe ratio of ≈ 0.72 across centuries of macroeconomic regimes.

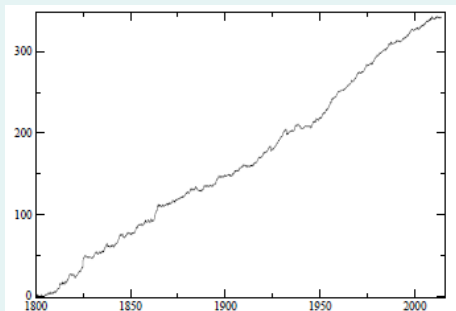


Figure: 200 Years of Trend Following performance across global asset classes (Lempérière et al. 2014).

Lempérière et al. prove that momentum is one of the rare systematic asset management strategies with intrinsic **positive return skewness** (right-skewed tail upside), contrasting with the severe negative skewness of carry and short volatility.

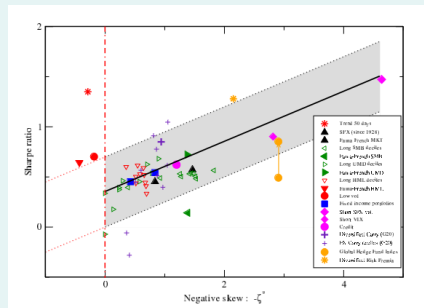


Figure: Left: Empirical right-skewed return distribution (Potters & Bouchaud). Right: Strategy Skewness vs. Sharpe Ratio across horizons (Lemprière et al.).

Factor Momentum (Ehsani & Linnainmaa 2022)

- **Core Finding: Individual stock momentum is largely inherited from momentum in underlying systematic factor returns**

$$(R_{i,t} = \sum \beta_{ik} F_{k,t} + \epsilon_{i,t}).$$
- Implication: Cross-sectional stock momentum disappears when controlling for factor persistence.

Table 6
Regressions of excess stock and bond returns (in percent) on the excess market returns (RM-RF) and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: July 1963 to December 1991, 342 months.^a

$$R(t) = \beta_F F(t) + \alpha + \beta_{RM}(t) - R_F(t) + \beta_{SMB}(t) + \beta_{HML}(t) + \epsilon(t)$$

Dependent variable: Excess returns on 25 stock portfolio formed on size and book-to-market equity

Size quintile	Book-to-market equity (BE/ME) quintiles					Book-to-market equity (BE/ME) quintiles				
	Low	2	3	4	High	Low	2	3	4	High
	<i>b</i>					<i>r(b)</i>				
Small	1.04	1.02	0.95	0.91	0.86	30.37	51.80	60.44	50.73	57.89
2	1.11	1.06	1.00	0.97	1.09	52.49	61.18	55.88	61.54	65.52
3	1.12	1.02	0.98	0.97	1.09	56.88	53.17	50.78	54.38	52.52
4	1.07	1.08	1.04	1.05	1.18	53.94	53.51	51.21	47.09	46.10
Big	0.96	1.02	0.98	0.99	1.06	60.93	56.76	46.57	53.87	38.61
	<i>s</i>					<i>t(s)</i>				
Small	1.46	1.26	1.19	1.17	1.23	37.92	44.11	52.03	52.85	50.97
2	1.00	0.98	0.88	0.73	0.89	32.73	38.79	34.03	31.66	36.78
3	0.76	0.65	0.60	0.48	0.66	26.40	23.39	21.71	18.67	21.91
4	0.37	0.33	0.29	0.24	0.41	12.73	11.11	9.81	7.38	11.01
Big	-0.17	-0.12	-0.23	-0.17	-0.05	-7.18	-4.51	-7.58	-6.27	-1.18
	<i>h</i>					<i>r(h)</i>				
Small	-0.29	0.08	0.26	0.40	0.62	-6.47	2.35	9.66	15.53	22.24
2	-0.52	0.01	0.26	0.46	0.70	-14.57	0.41	8.56	17.24	24.86
3	-0.34	0.00	0.32	0.51	0.68	-11.26	0.05	6.75	16.88	19.39
4	-0.42	0.04	0.30	0.56	0.74	-12.51	1.04	8.83	14.84	17.09
Big	-0.46	0.00	0.21	0.57	0.76	-17.03	0.09	5.80	18.34	16.24

E. F. Fama and K. R. French, Common risk factors in stock and bond returns

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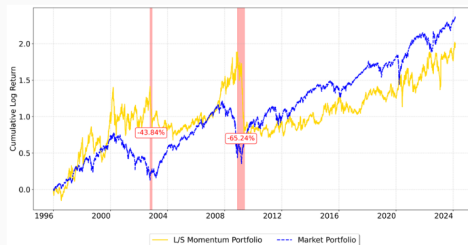
E. F. Fama

Momentum Crashes (Daniel & Moskowitz 2016)

The Rebound Crash Anomaly

Unhedged equity cross-sectional momentum (XSMOM) experiences catastrophic drawdowns during market V-rebounds following major panics (1932, 2008–2009 GFC) due to short-leg call-option squeezes on distressed "loser" stocks.

Figure 1: Cumulative Log Returns of L/S Momentum vs Global Equity Market



Source: Verdad analysis, Capital IQ

Figure: Empirical drawdown profile of Cross-Sectional Momentum (XSMOM) during the Global Financial Crisis (Daniel & Moskowitz 2016).

Crash Mitigation & Dynamic Risk Scaling

Target-Volatility Position Scaling

Position sizes must scale inversely with historical volatility: $w_{i,t} = \frac{\sigma_{\text{target}}}{\hat{\sigma}_{i,t}} \cdot s_{i,t}$.

- ▶ **Volatility Timing: Scaling leverage down during high-volatility panic regimes caps tail risk.**
- ▶ **Combined Filtering:** Combining volatility targeting with absolute trend filters (TSMOM) neutralizes short-leg squeeze risk during market turnarounds.

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Classical Filters & Lag Drag (Bruder & Gaussel 2011)

- ▶ **Simple Moving Average (SMA):** $SMA_t(k) = \frac{1}{k} \sum_{i=0}^{k-1} P_{t-i}$. **Suffers from step-response lag.**
- ▶ **Exponentially Weighted Moving Average (EWMA):**
 $EWMA_t(\lambda) = \lambda P_t + (1 - \lambda)EWMA_{t-1}(\lambda)$. Half-life $H = \ln(0.5)/\ln(1 - \lambda)$.
- ▶ **The Trade-Off Dilemma:** Short lookbacks increase false discovery from noise; long lookbacks suffer severe lag drag during trend reversals.

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Kalman Filtering & Bayesian Changepoint Detection

Local Linear Trend Model (State-Space)

$$y_t = L_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2) \quad (1)$$

$$L_t = L_{t-1} + T_{t-1} + \eta_{1,t}, \quad T_t = T_{t-1} + \eta_{2,t} \quad (2)$$

Estimates latent price level L_t and trend velocity T_t dynamically with time-varying parameters.

Bayesian Online Changepoint Detection (BOCPD)

Tracks real-time hazard rates to identify structural V-bottom reversals. Resets filter covariance matrices and scales down leverage during regime breaks.

Neural Momentum Architectures (Lim et al. 2019, Wood et al. 2021)

DMNs (LSTMs) & Momentum Transformers (TFTs)

Combines trend estimation and position sizing into end-to-end deep networks trained directly to maximize Sharpe ratio. Attention mechanisms capture multi-horizon temporal context.

Critical Practitioner Evaluation: The Drawbacks

- ▶ **Massive Data Requirements:** Deep architectures require immense sample size, making them ill-suited for low-frequency macro data.
- ▶ **High Sample Variance:** Training outputs exhibit extreme variance across random initialization seeds.
- ▶ **Overfitting Non-Stationary Regimes:** High risk of memorizing past regime noise rather than true structural trend persistence.

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The Martin-Zou Theorem (Martin & Zou 2012)

Martin-Zou Skewness Theorem

Filtering i.i.d. normal shocks $X_t \sim \mathcal{N}(0, \sigma^2)$ via an EWMA filter $s_t = c \sum_{k=1}^{\infty} \rho^k X_{t-k}$ generates positive strategy skewness over horizon T :

$$\mathbb{E}[R_{t+T}^3] = 6 \sum_{j=1}^T (T-j) a_j \gamma_j^3 > 0.$$

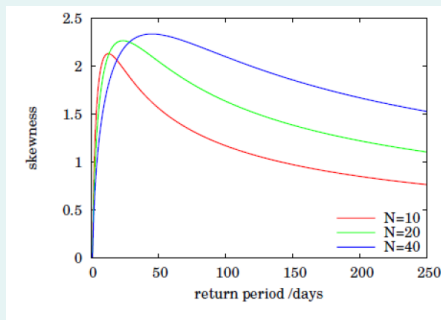


Figure: Analytical Skewness vs. Horizon curves generated by EWMA filters (Martin & Zou 2012).

Lookback Straddle Equivalence (Fung & Hsieh 2001)

Convexity & Option-Like Payoff Profile

Dynamic trend following generates a convex return profile mathematically equivalent to a lookback straddle:

$$\text{Payoff}_{\text{lookback}} = \max_{0 \leq t \leq T} P_t - \min_{0 \leq t \leq T} P_t \quad (3)$$

- **Theoretical Intuition: Captures extreme market moves in either direction, explaining positive return skewness during major trends.**
- **Practical Distinction:** Unconstrained lookback options are non-tradable; the equivalence provides structural intuition for gamma-like payoff convexity rather than a direct replication blueprint.

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Multi-Style Premia: Momentum, Carry & Value (Baz et al. 2015)

Combining Cross-Sectional & Time-Series Momentum

Baz, Rattray et al. demonstrate that combining both Cross-Sectional (XS) and Time-Series (TS) momentum with Carry and Value strategies exploits cross-style uncorrelation, dramatically enhancing risk-adjusted returns and drawdown control.

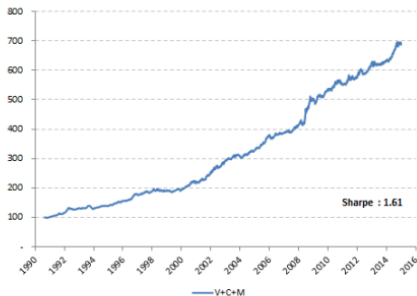


Figure 11. CS+TS: Value+Carry+Momentum Compounded P&L

Python Implementation: Volatility-Scaled Trend Engine

```
import numpy as np
import pandas as pd

class VolatilityScaledTSMOM:
    def __init__(self, fast=32, slow=96, vol_span=60, target_vol=0.15):
        self.fast, self.slow = fast, slow
        self.vol_span, self.target_vol = vol_span, target_vol

    def run(self, prices: pd.Series) -> dict:
        returns = prices.pct_change()
        fast_e = prices.ewm(span=self.fast, adjust=False).mean()
        slow_e = prices.ewm(span=self.slow, adjust=False).mean()
        signal = np.sign(fast_e - slow_e)

        ann_vol = (returns.ewm(span=self.vol_span, adjust=False).std() * np.sqrt(252)).clip(lower=0.01)
        positions = signal * (self.target_vol / ann_vol)
        strat_ret = positions.shift(1) * returns

        return {
            'sharpe': np.sqrt(252) * strat_ret.mean() / strat_ret.std(),
            'skewness': strat_ret.skew()
        }
```

Academic Literature & References

Primary Academic Bibliography (Clickable Live Links)

- ▶ Jegadeesh, N., & Titman, S. (1993). Returns to Buying Winners and Selling Losers. *Journal of Finance*.
- ▶ Fung, W., & Hsieh, D. A. (2001). The Risk in Trend-Following Strategies. *Review of Financial Studies*.
- ▶ Bouchaud, J.-P., & Potters, M. (2005). Momentum Trading: More Winners than Losers. *arXiv:physics/0508104*.
- ▶ Bruder, B., & Gaussel, N. (2011). Risk-Return Analysis of Dynamic Investment Strategies. *SSRN Working Paper*.
- ▶ Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time Series Momentum. *Journal of Financial Economics*.
- ▶ Martin, R., & Zou, D. (2012). Momentum Trading 'Skews Me. *Risk Magazine*.
- ▶ Hamdan, R., Pavlowsky, F., Roncalli, T., & Zheng, B. (2012). Understanding the Momentum Risk Premia. *Lyxor Asset Management Working Paper*.
- ▶ Asness, C. S., Moskowitz, T. J., & Pedersen, L. H. (2013). Value and Momentum Everywhere. *Journal of Finance*.
- ▶ Lempérière, G., et al. (2014). Two Centuries of Trend Following. *Journal of Investment Strategies*.
- ▶ Baz, J., Rattray, M., Harvey, C. R., Van Hemert, O., & Hofmann, G. (2015). Dissecting Investment Strategies: Momentum, Value, Carry. *Man AHL / SSRN Working Paper*.
- ▶ Daniel, K., & Moskowitz, T. J. (2016). Momentum Crashes. *Journal of Financial Economics*.
- ▶ Lempérière, G., et al. (2017). Risk Premia: Asymmetric Tail Risks. *Quantitative Finance*.
- ▶ Lim, B., Zohren, S., & Roberts, S. (2019). Deep Momentum Networks. *Journal of Financial Data Science*.
- ▶ Ehsani, S., & Linnainmaa, J. T. (2022). Factor Momentum and the Momentum Factor. *Journal of Finance*.
- ▶ Wood, K., Roberts, S., & Zohren, S. (2021). Slow Momentum with Fast Reversion: A Trading Strategy Using Deep Learning and Changepoint Detection. *Journal of Financial Data Science*.
- ▶ Wood, K., Giegerich, S., Roberts, S., & Zohren, S. (2021). Trading with the Momentum Transformer: An Intelligent and Interpretable Architecture. *arXiv:2112.08534*.